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Summary Sheet

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Segment-Coupled Gaussian and Advection–Diffusion-Based Multi-Scale Urban Pollutant Transport Model

Summary

Urban air pollution represents a significant environmental issue affecting public health, and the mechanisms governing its transport from outdoor sources into indoor spaces are highly complex. In this study, a multi-scale mathematical model is developed to simulate the entire process of pollutant movement in complex urban environments, encompassing both "outdoor dispersion–indoor distribution," with precise spatiotemporal resolution for sudden atmospheric pollution events.

For Problem 1, a **Segment-Coupled Gaussian Plume Model** is established to simulate pollutant dispersion over heterogeneous underlying surfaces. Considering abrupt changes in turbulence characteristics caused by the interspersed arrangement of buildings, water bodies, and green spaces in urban environments, the model incorporates a **Virtual Source Method** to adjust dispersion parameters along the transport path, achieving smooth concentration transitions across regions. On this basis, the model integrates gravity settling corrections and dry deposition mechanisms, and introduces a **ground reflection coefficient** according to different surface characteristics to enhance the physical realism of particulate deposition simulations. Under extreme calm wind conditions ($u < 0.5$ m/s), the model automatically switches to a **Gaussian Puff Model**, addressing the singularity issues present in conventional models at low wind speeds. Simulation results using the Shanghai Oriental Pearl Tower and the Canton Tower as emission sources indicate that pollutants primarily form high-concentration bands along the downwind axis, with concentrations decaying nonlinearly with deviations between receptor and emission heights. In the far-field lateral direction, pollutant concentrations rapidly dilute to negligible levels.

For Problem 2, to analyze the dynamic transmission of outdoor pollutants into indoor environments, a **three-dimensional advection–diffusion-based indoor air quality model** is developed. This model comprehensively accounts for three primary transport pathways: infiltration through door and window cracks, natural ventilation, and mechanical ventilation. Corresponding boundary conditions are constructed, including crack boundaries with infiltration coefficients and ventilation openings based on flow-field fluxes. The governing equations are solved numerically using the **finite difference method**. Temporal and spatial simulations are performed under eight combined scenarios, showing that convective effects induced by natural ventilation dominate pollutant dispersion, whereas pollutant transport through crack infiltration alone exhibits significant time-lag effects. Based on these findings, an emergency strategy is proposed, prioritizing the closure of natural ventilation systems and the use of high air-tightness rooms for **shelter-in-place** during sudden pollution events.

In summary, the modeling framework developed in this study enables end-to-end simulation from the urban scale down to the building scale, providing a quantitative theoretical basis for emergency response and building protection during sudden pollution events.

Keywords: Gaussian Plume Model; Virtual Source Method; Advection–Diffusion Equation; Finite Difference Method; Indoor Air Quality

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1. Introduction

1.1. Problem Background

With rapid urbanization and industrialization in China, air pollution has become a major challenge to public health and quality of life[1]. Emissions from industrial activities and traffic, combined with complex urban meteorological and spatial conditions, often lead to regional air pollution that poses significant health risks. Effective pollution control therefore requires a clear understanding of pollutant emission and dispersion processes[2].



Figure 1: Air Pollution Map of Shanghai Residential Areas

By integrating modern monitoring technologies and numerical simulation, dynamic models can be developed to assess pollution levels and support optimized air quality management[3]. Consequently, studying pollutant emission and dispersion mechanisms is essential for improving air quality, protecting public health, and promoting sustainable urban development.

1.2. Restatement of the Problem

Based on the context of urban air pollution and potential large-scale emissions, our objective is to develop rigorous mathematical models to assess the impact of pollutant sources on residential air quality. The models should address the following issues:

1.2.1. *Problem 1: Outdoor Air Pollution Impact on Residential Areas*

- **Source Location and Emission Height:** Investigate how the height of a pollutant source and its distance from residential areas influence pollutant concentrations in the surrounding neighborhoods.
- **Large-Scale Pollutant Release Scenario:** Assume a significant release of pollutants occurs. Use Guangzhou's Haixin Tower as a representative case study to quantify the impact on nearby residential areas.
- **Surface Type Effects:** Evaluate how different underlying surface types (e.g., bare ground, vegetation, and building coverage) affect pollutant dispersion and regional exposure patterns.

1.2.2. *Problem 2: Outdoor-to-Indoor Pollution Transmission*

- **Indoor Air Quality Implications:** Analyze the influence of varying outdoor pollutant concentrations, including those derived from Problem 1, on indoor air quality in residential spaces.
- **Transmission Pathways:** Consider three primary mechanisms for outdoor pollutants entering indoor environments: infiltration through doors and windows, natural ventilation, and mechanical ventilation systems.
- **Dynamic Modeling:** Develop a dynamic mathematical model to describe the transport and diffusion of outdoor pollutants into indoor spaces, and to quantify resultant indoor pollutant concentrations under different outdoor pollution scenarios.

1.3. Our Work

To avoid complicated description and intuitively reflect our work process, the flowhart is shown as the following Figure 2Figure 3.

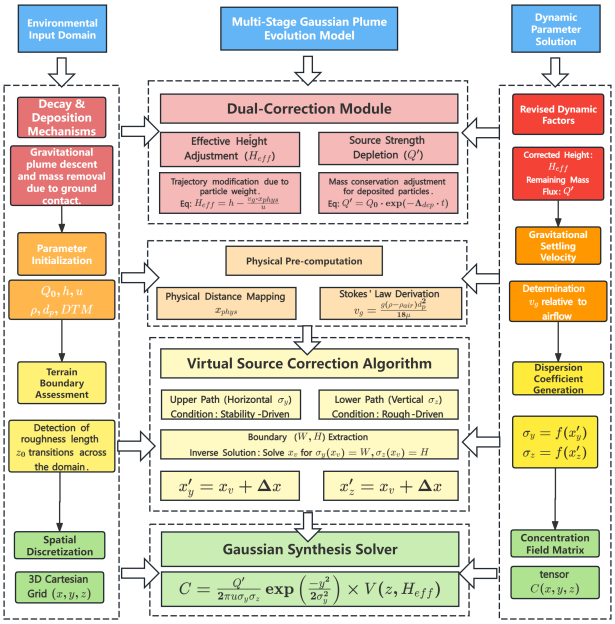


Figure 2: problem 1

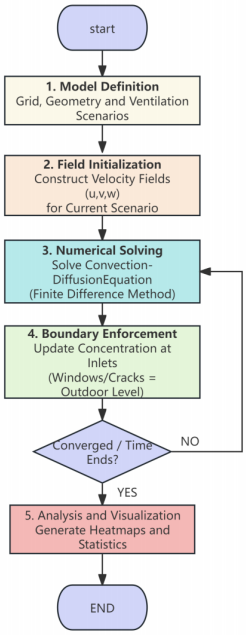


Figure 3: problem 2

2. Model Preparations

2.1. Assumptions and Justifications

Before constructing our model, we establish the following assumptions based on physical principles and practical considerations:

Assumption 1: Pollutants consist of fine spherical particles, and mass is conserved during transport.

Justification: Fine particles follow airflow and can be treated as spheres; mass conservation ensures model consistency, while short-term transport allows neglect of reactions and long-range deposition.

Assumption 2: Wind and turbulence regimes are simplified depending on wind speed.

Justification: Wind is quasi-steady; advection dominates at high speeds and turbulence at low speeds, enabling tractable modeling under different conditions.

Assumption 3: Terrain is effectively flat, with surface effects captured via empirical dispersion coefficients.

Justification: Flat terrain standardizes Gaussian models; surface type effects are included via dispersion parameters and reflection coefficients.

Assumption 4: Pollutant concentrations follow Gaussian distributions in crosswind and vertical directions, both outdoors and indoors.

Justification: Gaussian distributions are validated for point sources; indoor air is assumed well-mixed with linear superposition of ventilation contributions.

Assumption 5: Outdoor pollutant concentrations are quasi-steady within each time step, and penetration through building envelopes is approximated with constant parameters.

Justification: Quasi-steady assumption allows combined analytical/numerical solutions; envelope penetration, filtration, and deposition are treated as constants for simplicity.

2.2. Notations

Table 1: Model Parameters and Symbol Definitions

Sym.	Definition	Unit	Sym.	Definition	Unit
Indoor IAQ & Basic Variables					
V	Indoor volume	m^3	C_i	Ind. poll. conc.	$\mathrm{mg/m}^3$
C_o	Out. poll. conc.	$\mathrm{mg/m}^3$	Q_t	Tot. vent. rate	m^3/s
k	Decay coeff.	s^{-1}	S	Source strength	$\mathrm{mg/s}$
A_g	Crack area	m^2	C_d	Crack flow coeff.	–
ΔP	Press. diff.	Pa	n	Flow exponent	–
A_e	Opening area	m^2	Δh	Height diff.	m
C_w	Wind coeff.	–	Q_d	Design vent. rate	m^3/s
η	Filt. efficiency	–	x, y, z	Coordinates	m
C	Poll. conc.	$\mathrm{mg/m}^3$	Q	Source strength	$\mathrm{mg/s}$
u	Avg. wind speed	$\mathrm{m/s}$			
Atmospheric Dispersion Parameters					
D_y	y-diff. coeff.	m^2/s	D_z	z-diff. coeff.	m^2/s
H_e	Effective height	m	α	Refl. coeff.	–
v_s	Sedim. velocity	$\mathrm{m/s}$	v_d	Dry depo. vel.	$\mathrm{m/s}$
u_e	Eff. wind speed	$\mathrm{m/s}$	σ	Turb. diff. vel.	$\mathrm{m/s}$
g	Grav. accel.	$\mathrm{m/s}^2$	ρ_p	Particle density	$\mathrm{kg/m}^3$

Sym.	Definition	Unit	Sym.	Definition	Unit
ρ_a	Air density	$\mathrm{kg/m^3}$	d_p	Particle diam.	m
μ	Dynamic visc.	$\mathrm{Pa\,s}$	Q'	Atten. source	$\mathrm{mg/s}$
u_m	Meas. wind sp.	$\mathrm{m/s}$	p	Wind profile exp.	–
Stab.	Stability class	–			

3. Problem 1 Model

3.1. Coordinate System Establishment and Simplification

Atmospheric pollutant dispersion obeys mass conservation and the convection-diffusion equation. Under steady, uniform wind with stationary turbulence, neglecting reactions and phase changes, an analytical solution can be derived. Based on gradient transport and stochastic turbulence, this yields the Gaussian Plume Model, which is concise and physically intuitive, suitable for mid-to-far field regions.

Basic Model Assumptions: The pollutant source is released continuously and stably. The underlying surface is flat and uniform. The average wind speed is spatially constant. Turbulence is homogeneous and isotropic. Concentration distributions in the crosswind and vertical directions follow a Gaussian approximation.

A right-handed Cartesian coordinate system is established with the pollution source as the origin: the x-axis follows the wind direction, y is the crosswind direction, and z is vertical (where H is the chimney or source emission height). Based on Fick's Law and mass conservation, the concentration C at any point (x, y, z) is expressed as:

$$C(x,y,z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)\right] \tag{1}$$

Lateral dispersion is controlled by σ_y, which increases with downwind distance. The two terms within the brackets represent the direct plume and the contribution from the mirror source, respectively, ensuring zero flux penetration at the ground.

3.2. Virtual Source Coupling Model for Non-uniform Surfaces

In urban environments, pollutants disperse across varying underlying surfaces (buildings, green spaces, water), each with different roughness and turbulent characteristics. This causes spatial variations in dispersion parameters. While the Gaussian assumption holds within each zone, standard models struggle with abrupt boundaries. Thus, a **Virtual Source Method** is introduced to construct a segmented coupling model.

Assume pollutants disperse in Region I ($0 \le x \le x_1$) before entering Region II ($x > x_1$). At the interface x_1, to avoid non-physical jumps in dispersion parameters, the virtual source position x_{virtual} is calculated so that the starting dispersion in Region II matches the end of Region I:

$$\sigma_{z,\mathrm{I}}(x_1) = \sigma_{z,\mathrm{II}}(x_{\text{virtual}}) \tag{2}$$

For $x > x_1$, the actual dispersion parameter is $\sigma_z(x) = \sigma_{z,\mathrm{II}}(x - x_1 + x_{\text{virtual}})$. The crosswind parameter σ_y follows the same logic. This method smooths parameter transitions and can be applied repeatedly across multiple regions.

3.3. Gravitational Settling, Dry Deposition, and Surface Reflection

In real-world environments, pollutants (especially particulates) are influenced by gravity, ground adsorption, and obstacle rebound.

3.3.1. *Gravitational Settling Correction*

Particulates undergo downward movement due to gravity, causing the plume center to shift vertically. Assuming a concentrated particle size distribution, the settling is characterized by a settling velocity v_s.

The transport time to a downwind distance x is $t = x/u$. The resulting downward displacement is $v_g t$, and the effective emission height becomes:

$$H_{\text{eff}}(x) = H - \frac{v_g x}{u} \tag{3}$$

The terminal settling velocity v_g in still air is given by Stokes Law:

$$v_g = \frac{\rho_p d_p^2 g}{18\mu} C_c \tag{4}$$

Substituting H_{eff} into the Gaussian formula accounts for settling while maintaining the model's analytical structure:

$$C(x,y,z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z - H_{\text{eff}})^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z + H_{\text{eff}})^2}{2\sigma_z^2}\right)\right] \tag{5}$$

This correction method incorporates the deposition effect into the model in the form of an effective height without altering the Gaussian dispersion structure, ensuring both the continuity of the model formulation and physical consistency.

3.3.2. *Ground Rebound and Reflection*

Particulates hitting the ground do not always reflect completely like gases. This is characterized by a reflection coefficient α. The corrected formula is:

$$C(x,y,z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-H_{\text{eff}})^2}{2\sigma_z^2}\right) + \alpha \exp\left(-\frac{(z+H_{\text{eff}})^2}{2\sigma_z^2}\right)\right] \tag{6}$$

The second term represents the vertical reflection correction introduced by the reflection coefficient. When $\alpha = 1$, the model reduces to the case of complete reflection; when $\alpha < 1$, it indicates that the ground partially adsorbs or removes pollutants.

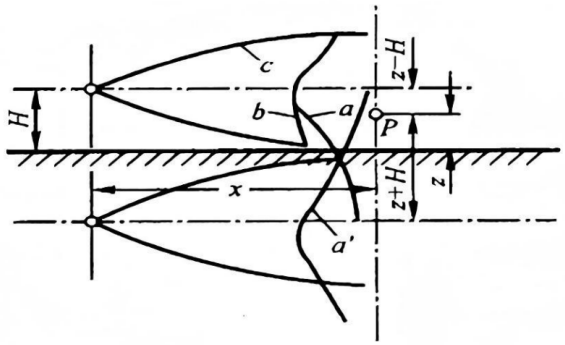


Figure 4: Ground Reflection Model

The reflection capability of pollutants varies significantly across different types of underlying surfaces. In practical calculations, the reflection coefficient α is chosen according to the type of underlying surface at the receptor location. Typical values of the reflection coefficient for various surface types are listed in Table 2.

Table 2: Ground Reflection Coefficients for Different Surfaces α

Underlying Surface Type	**Reflection Coefficient** (α)	**Description**
Concrete, Asphalt	0.9 – 1.0	Near-total reflection
Water Surface	0.9 – 1.0	Basic total reflection, possible slight absorption
Grass, Low Vegetation	0.6 – 0.8	Partial adsorption
Shrubs	0.4 – 0.6	Significant adsorption
Dense Forest	0.2 – 0.4	Strong adsorption

3.3.3. *Dry Deposition Correction*

Surfaces like vegetation and water adsorb particulates, causing the effective source strength to decay with distance. The decayed source strength $Q'(x)$ is:

$$Q'(x) = Q \cdot \exp\left(-\frac{\sqrt{2/\pi}\, v_d\, (x - x_1)}{u\, \sigma_{z,\text{avg}}}\right) \tag{7}$$

Here, $\sigma_{z,\text{avg}}$ is the average vertical dispersion parameter in Region 2, and v_d is the dry deposition velocity. Reference values are approximately 0.001 m/s for PM2.5, 0.005 m/s for PM10, and 0.008 m/s

for gaseous pollutants such as SO_2. Dry deposition velocities vary across different underlying surfaces, e.g., 0.1–0.3 cm/s over water and 0.5–2.0 cm/s over vegetation.

Considering both gravitational settling and dry deposition corrections, the complete Gaussian dispersion formula is given by:

$$C(x, y, z) = \frac{Q'(x)}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \cdot C_z(z, H, v_s, x) \tag{8}$$

3.4. Determination of Key Model Parameters

In atmospheric dispersion simulations, model accuracy depends on key physical parameters, including source emission height, atmospheric stability, and corresponding dispersion coefficients. A virtual source-based path correction is also applied to account for complex underlying surfaces.

3.4.1. *Effective Emission Height*

The effective emission height H, which determines the plume centerline, is $H = h_s + \Delta h$, where h_s is stack height and Δh is plume rise due to emission momentum and thermal buoyancy. In cold releases, $\Delta h = 0$, and this study focuses on such scenarios for conservative environmental risk assessment.

3.4.2. *Atmospheric Stability and Dispersion Parameters*

Atmospheric stability directly characterizes the intensity of thermal and mechanical turbulence and is a key factor controlling the rate of pollutant dispersion. This model follows the Pasquill-Gifford classification, which categorizes atmospheric conditions into six classes, A through F[4].

Table 3: Atmospheric Stability Classifications

Level	**Description**	**Characteristics**	**Example Scenario**
A	Extremely Unstable	Strong thermal convection	Sunny summer noon, low wind (<3m/s)
B	Unstable	Moderate convection	Sunny day, moderate wind
C	Slightly Unstable	Weak convection	Cloudy day
D	Neutral	Mechanical turbulence	Overcast or high wind (>6m/s)
E	Slightly Stable	Initial inversion	Night, few clouds, low wind
F	Stable	Strong inversion	Night, clear, calm

For complex urban environments, this study adopts the urban dispersion parameterization scheme proposed by Briggs. In this scheme, the lateral and vertical dispersion parameters, σ_y and σ_z, are expressed as functions of the downwind distance x, capturing the distinct evolution characteristics of the plume in the near field and far field:

$$\sigma_y(x) = a_y x(1 + b_y x)^{c_y} \tag{4.3.1}$$

$$\sigma_z(x) = a_z x(1 + b_z x)^{c_z} \tag{4.3.2}$$

Near the source (x small), dispersion parameters grow approximately linearly, reflecting initial plume expansion, while growth slows in the far field due to large-scale turbulence. Under unstable conditions (A–B), coefficients a are larger than in stable conditions (E–F), explaining faster dilution in convective weather and elongated high-concentration plumes under inversions. Coefficient values are given in the table below[4].

Table 4: Briggs Urban Dispersion Coefficients

Stability Class	$\sigma_y(x)$	$\sigma_z(x)$
A-B	$0.32x(1+0.0004x)^{-0.5}$	$0.24x(1+0.001x)^{0.5}$
C	$0.22x(1+0.0004x)^{-0.5}$	$0.20x$
D	$0.16x(1+0.0004x)^{-0.5}$	$0.14x(1+0.0003x)^{-0.5}$
E-F	$0.11x(1+0.0004x)^{-0.5}$	$0.08x(1+0.00015x)^{-0.5}$

When pollutants move across surfaces with differing roughness and thermal properties (e.g., urban to water), a single dispersion model is insufficient. The model switches from "Briggs urban" to "Pasquill-Gifford rural," with the virtual source method ensuring continuous σ values, realistically capturing narrow high-concentration bands over water.

3.4.3. *Wind Speed Profile Correction*

Wind speed is corrected to the emission height H using a power-law profile:

$$u(H) = u_{10}\left(\frac{H}{10}\right)^{p} \tag{9}$$

In this process, the wind profile exponent p reflects the influence of surface roughness. For multi-region transport paths, the model employs a distance-based weighted correction method. At points where $x > x_1$, the effective exponent is calculated as

$$p_{\text{eff}}(x) = \frac{p_{\text{I}}x_1 + p_{\text{II}}(x - x_1)}{x}, \tag{10}$$

where p_{I} and p_{II} are the roughness exponents for the two respective regions.

Table 5: Roughness Index p for Different Surfaces

Underlying Surface Type	**Roughness Length z_0 (m)**	**Index p**	**Typical Scenario**
Open Water	0.0001 - 0.001	0.10 - 0.15	Pearl River, Huangpu River
Flat Open Land	0.01 - 0.05	0.15 - 0.20	Airport, Grassland
Low Vegetation	0.1 - 0.3	0.20 - 0.25	Suburban Farmland
Urban / Forest	0.5 - 1.0	0.25 - 0.30	General City Area
Dense High-rise	1.0 - 2.0	0.30 - 0.40	Lujiazui, Zhujiang New Town

3.4.4. *Normalization of Initial Source Strength*

Due to the difficulty in accurately predicting accidental source strength, Q is normalized as $Q_{\text{norm}} = 1000\ \text{mg} \cdot \text{s}^{-1}$. Simulated concentrations are relative and can be scaled to actual conditions using:

$$C_{\text{actual}} = C_{\text{simulated}} \times \frac{Q_{\text{actual}}}{Q_{\text{norm}}}. \tag{11}$$

3.5. Complete Formulation of the Segment-Coupled Model

The segment-coupled Gaussian model first identifies the receptor's region. For points in Region II, where surface properties change, the virtual source method calculates x_{virtual} for smooth dispersion parameter transitions, and a weighted roughness exponent p_{eff} accounts for path effects. The effective source strength Q' is updated via dry deposition correction, and concentrations are computed using corrected parameters and wind. This ensures continuity at boundaries and captures spatial variations in wind, ground reflection, and deposition, improving accuracy in complex environments.

Concentration distribution in Region I ($x \leq x_1$):

$$C(x,y,z) = \frac{Q}{2\pi u_{\text{I}} \sigma_{y,\text{I}} \sigma_{z,\text{I}}} \exp\left(-\frac{y^2}{2\sigma_{y,\text{I}}^2}\right)\left[\exp\left(-\frac{(z-H)^2}{2\sigma_{z,\text{I}}^2}\right) + \alpha_{\text{I}} \cdot \exp\left(-\frac{(z+H)^2}{2\sigma_{z,\text{I}}^2}\right)\right] \tag{4.4.4}$$

Concentration distribution in Region II ($x > x_1$):

$$C(x,y,z) = \frac{Q'(x)}{2\pi u_{\text{II}} \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right)\left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \alpha_{\text{II}} \cdot \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)\right] \tag{4.4.5}$$

In this formulation, σ_y and σ_z are strictly evolved and corrected based on the virtual source distance. The transport wind speed in Region II, u_{II}, is determined by the path-weighted exponent $p_{\text{eff}}(x)$, while the attenuated effective source strength $Q'(x)$ is calculated using the average vertical dispersion parameter $\overline{\sigma}_z$ in Region II.

3.6. Model Implementation and Computation

After setting the segment-coupled Gaussian model parameters, pollutant dispersion was simulated under default meteorological conditions and the underlying surfaces. Analysis proceeds from high to low dimensionality—3D spatial distribution, 2D cross-sections, and 1D variable responses—to comprehensively reveal dispersion patterns.

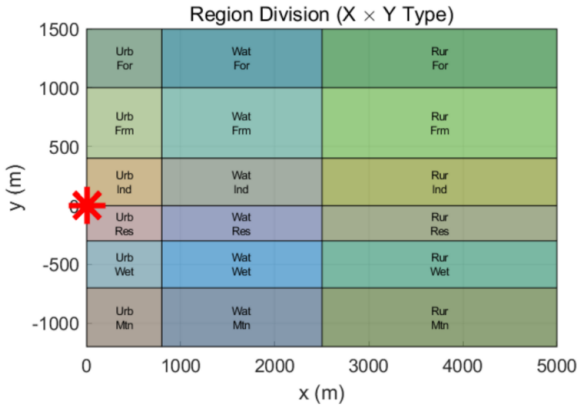


Figure 5: Illustration of the Gaussian Plume Model

Table 6:

Direction	**Type of Underlying Surface**	**Parameters** (α, p)
X	Urban	0.90, 0.28
X	Water	0.40, 0.11
X	Rural	0.65, 0.15
Y^+	Industrial	0.85, 0.25
Y^+	Farmland	0.70, 0.14
Y^+	Forest	0.55, 0.20
Y^-	Residential	0.88, 0.22
Y^-	Wetland	0.35, 0.10
Y^-	Mountain	0.50, 0.30

$x \in [800, 2500], \mathrm{m} y^+ \in [400, 1000], \mathrm{m} y^- \in [-300, -700], \mathrm{m}$

3.6.1. *Overall Ground-Level Pollutant Concentration (Three-Dimensional)*

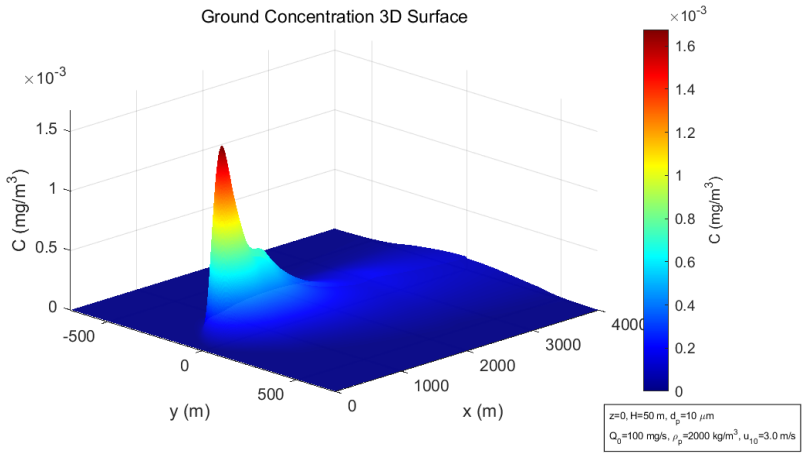


Figure 6: 3D Spatial Distribution of Overall Concentration

The figure shows the 3D concentration distribution of pollutants. The plume, originating from the source, exhibits a typical Gaussian shape: mainly transported along x under wind, while gradually dispersing in y and z. High concentrations occur near the downwind centerline around the emission height H, consistent with the Gaussian assumption.

Due to superposition of x, y, and z, directional dispersion and decay patterns are visually obscured. Dimensionality reduction to 2D and 1D analyses is therefore needed to reveal the underlying dispersion mechanisms.

3.6.2. *2D Cross-Section Concentration Heatmap Analysis*

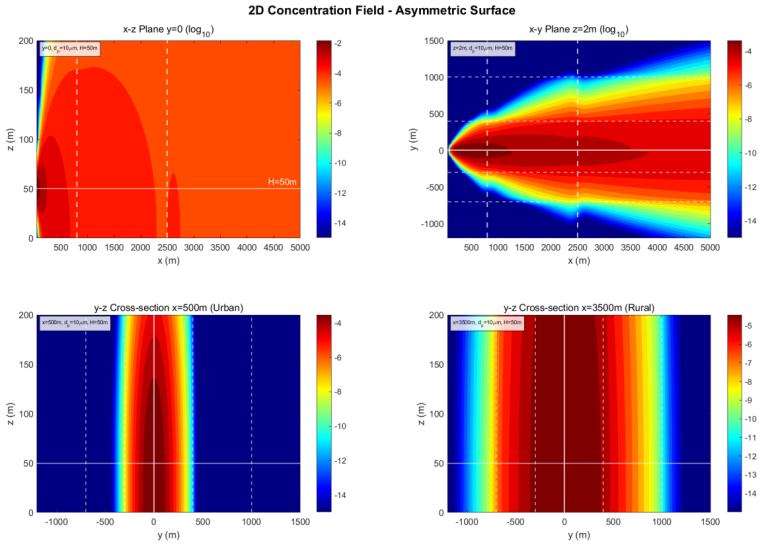


Figure 7: 2D Concentration Cross-Section Heatmap

The figure shows a 2D concentration heatmap along a selected plane, typically a longitudinal plane along the main wind axis or a near-ground horizontal plane, illustrating pollutant distribution. Concentrations are highest near the source and downwind centerline, especially around the emission height $H = 50$ m, forming a distinct high-concentration band. With increasing downwind distance x, the plume spreads laterally and vertically while peak concentrations decrease, reflecting mass conservation: as dispersion volume grows, pollutant mass per unit volume decreases.

3.6.3. *Lateral (y) and Vertical (z) Concentration Distribution*

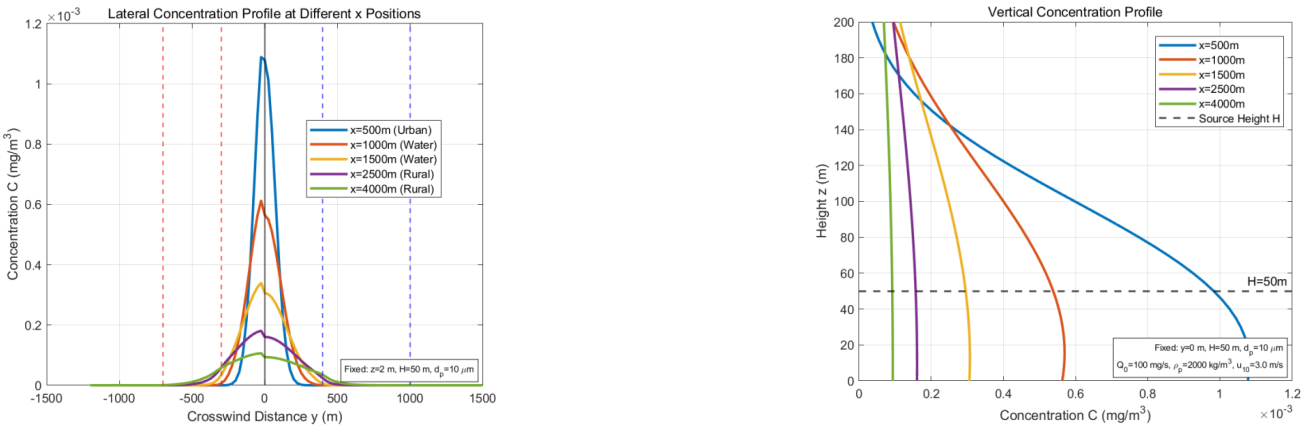


Figure 8: Concentration Distribution in the Lateral and Vertical Directions

In both lateral (y) and vertical (z) directions, concentrations exhibit Gaussian characteristics, widening and decreasing with downwind distance x. Laterally, turbulent diffusion spreads pollutants on both sides of the wind axis, while gravitational settling lowers the plume centerline, enhancing near-ground dispersion. Vertically, high concentrations are near the emission height H in the near field, but shift downward in the far field due to vertical dispersion and settling, reflecting the combined effects of advection, diffusion, and settling.

3.6.4. *Sensitivity Analysis under One-Dimensional Variables*

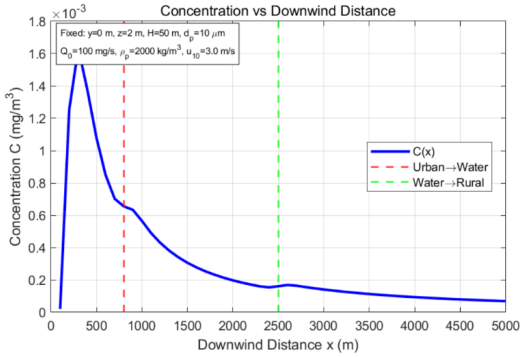


Figure 9: Relationship between pollutant concentration and downwind distance.

As downwind distance x increases, the plume broadens laterally (y), evolving from a narrow, high peak to a wide, low Gaussian profile. This results from cumulative turbulent diffusion increasing σ_y, while gravitational settling lowers the plume centerline, enhancing near-ground lateral spread and extending the plume's effective influence.

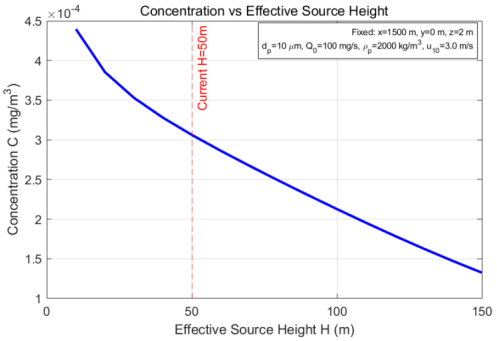


Figure 10: Relationship between pollutant concentration and source emission height.

At a fixed downwind distance ($x = 1500$ m), ground-level concentrations decrease roughly linearly with increasing emission height H, as higher releases allow more vertical dispersion and settling. While raising H reduces near-ground pollution, it may broaden the affected area, requiring careful engineering consideration.

3.7. Calm Wind Conditions

3.7.1. *Gaussian Puff Model*

Under extremely low wind speeds ($u < 0.5$ m/s), the conventional Gaussian plume model becomes singular and advection is no longer dominant. Pollutant dispersion shifts to primarily isotropic diffusion, requiring the Gaussian puff model and its steady-state solution. Pollutants disperse radially rather than forming elongated plumes. The calm-wind correction also includes effective emission height H_{eff}, gravitational settling, and a surface reflection coefficient α, leading to the steady-state concentration:

$$C(r,z) = \frac{Q'}{(2\pi)^{1.5}\sigma_r^2\sigma_z} \exp\left(-\frac{r^2}{2\sigma_r^2}\right)\left[\exp\left(-\frac{(z-H_{\text{eff}})^2}{2\sigma_z^2}\right) + \alpha \cdot \exp\left(-\frac{(z+H_{\text{eff}})^2}{2\sigma_z^2}\right)\right] \quad (4.4.6)$$

Here, $r = \sqrt{x^2+y^2}$ represents the radial distance between the receptor point and the emission source in the horizontal plane. Due to the isotropic nature of calm wind conditions, the horizontal dispersion parameter is set as $\sigma_r = \sigma_y$. In the absence of significant wind direction, the pollutant distribution exhibits high geometric symmetry, with the maximum ground-level concentration located directly beneath the source.

3.7.2. *Model Implementation*

After setting the parameters for the segment-coupled Gaussian puff model, numerical simulations of pollutant dispersion were carried out under default meteorological conditions and the underlying surface configurations described in figure13 (left). Lateral distance (y): 0 m (directly along the source centerline). Receptor height (z): 2 m (near-ground level). Effective emission height (H): 50 m. Particle diameter (d_p): 10 μm. Meteorological conditions: calm/no wind.

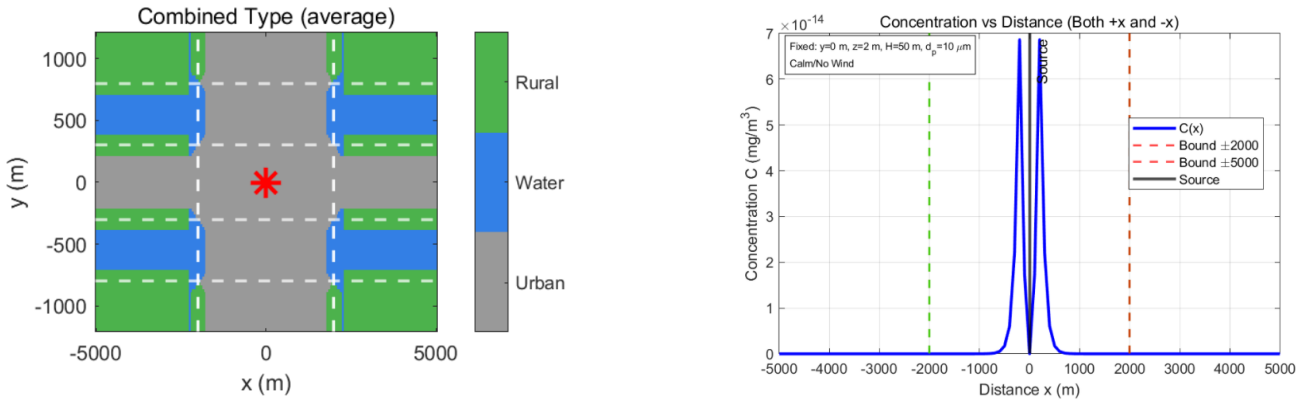


Figure 11

Figure 13 (right) shows pollutant concentrations along the x-axis under calm wind. Without advection, dispersion relies on turbulence and molecular diffusion, forming a bell-shaped profile centered at the source. Concentrations peak near $x = 0$ and decay exponentially with distance, indicating highly localized hazards and highlighting the need for a circular, source-centered alert zone in calm-wind scenarios.

3.8. Case Studies

To validate the segment-coupled Gaussian model for complex urban environments and accidental pollutant releases, this chapter simulates two megacities: Shanghai (Oriental Pearl Tower area) and Guangzhou (Canton Tower area).

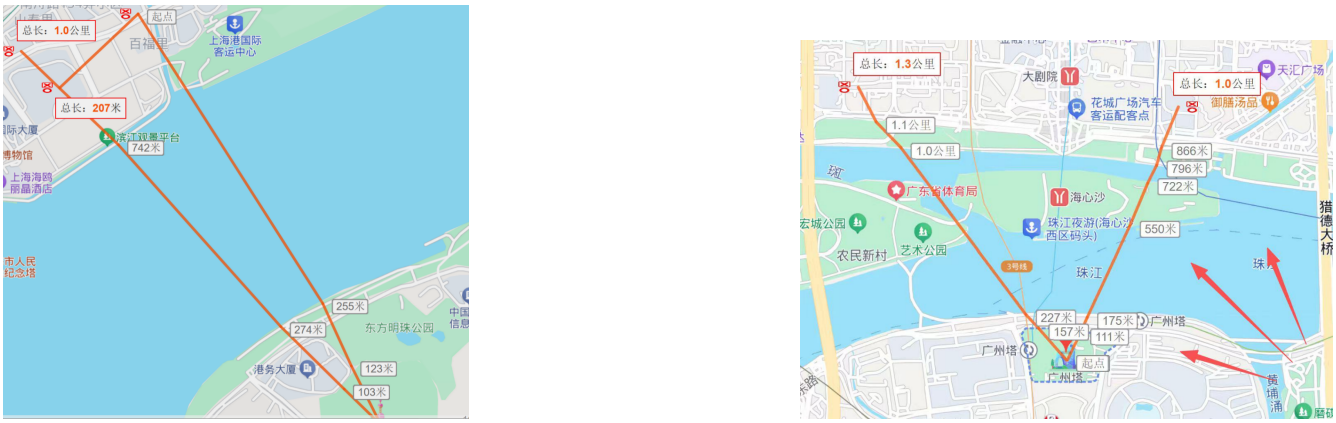


Figure 12: Underlying surface maps of Shanghai (left) and Guangzhou (right) residential areas.

Simulations over the complex terrain along the Huangpu River compare two Shanghai neighborhoods located on the downwind axis ($x = 1000$ m, $y = 0$ m) and at the lateral edge ($x = 915$ m, $y = 200$ m), revealing microscale differences in pollutant concentration. A parallel analysis in Guangzhou's Zhujiang New Town examines far-field dispersion by contrasting an axial neighborhood ($x = 1207$ m, $y = 0$ m) with a distant sidewind site ($x = 1000$ m, $y = 908$ m), where the larger spatial extent leads to more pronounced spatial heterogeneity.

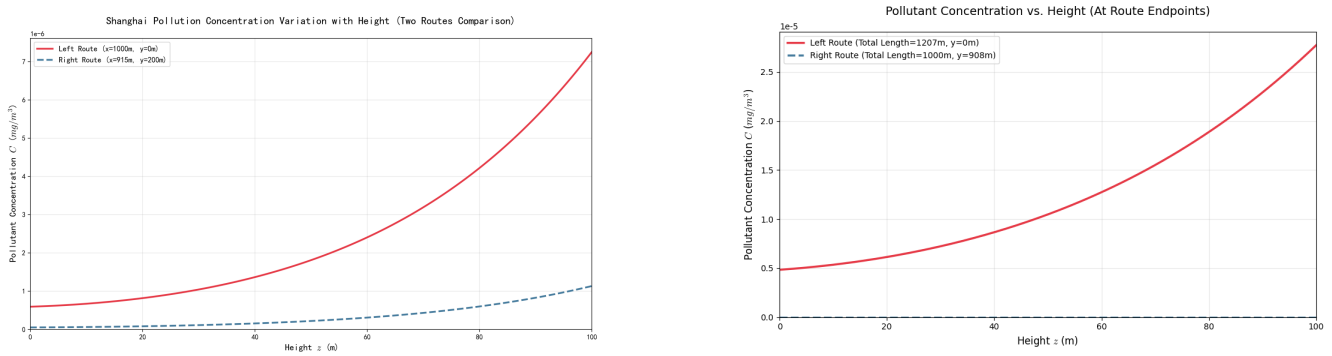


Figure 13: concentration comparison of two Shanghai and Guangzhou neighborhoods.

The Shanghai results indicate that concentrations along the downwind axis are significantly higher than those in the sidewind neighborhood, confirming the plume centerline as the core pollution zone with rapid lateral dilution, while concentrations in both neighborhoods increase with receptor height z as high-rise buildings approach the plume center height H. In contrast, the Guangzhou results show nearly zero concentrations in the sidewind far-field neighborhood due to strong dilution from lateral displacement and long transport distance, whereas the axial neighborhood retains noticeable concentrations that increase with height z, reflecting strong near-ground attenuation by deposition and the persistence of the plume aloft due to vertical diffusion.

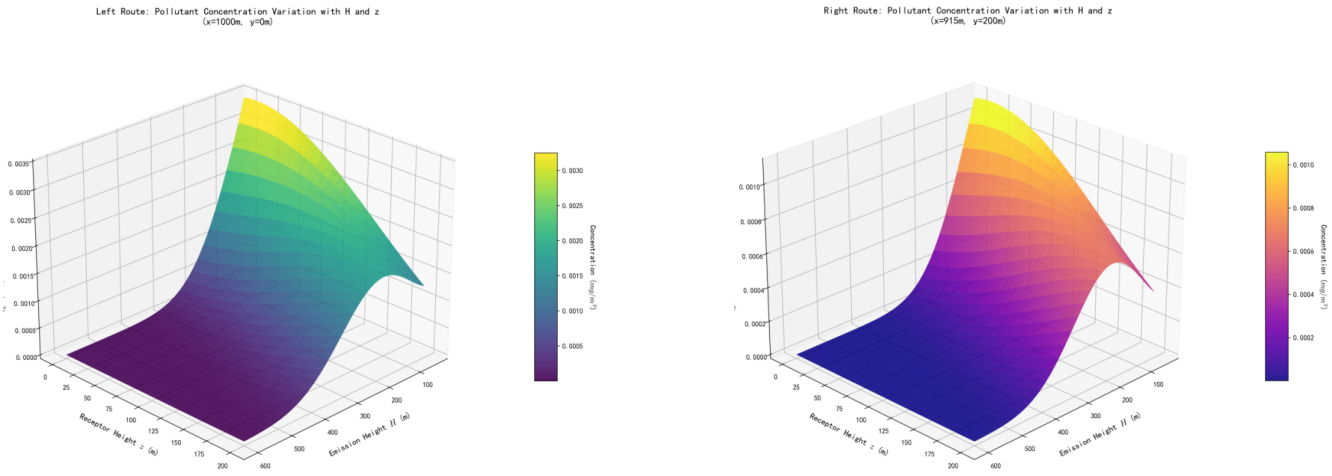


Figure 14: Three-dimensional concentration analysis of Shanghai neighborhoods.

The left neighborhood shows significantly higher pollutant concentrations than the right one, identifying the downwind axis ($x = 1000$ m, $y = 0$ m) as the core pollution region, while the sidewind neighborhood ($x = 915$ m, $y = 200$ m) experiences strong lateral dilution. In the far field, concentrations in the sidewind neighborhood are nearly zero due to lateral displacement and long transport distance, whereas the axial neighborhood retains residual concentrations that increase with receptor height z. Peak concentrations occur when the emission height H matches z, while deviations between them lead to rapid attenuation, confirming the nonlinear spatial response of the Gaussian dispersion model.

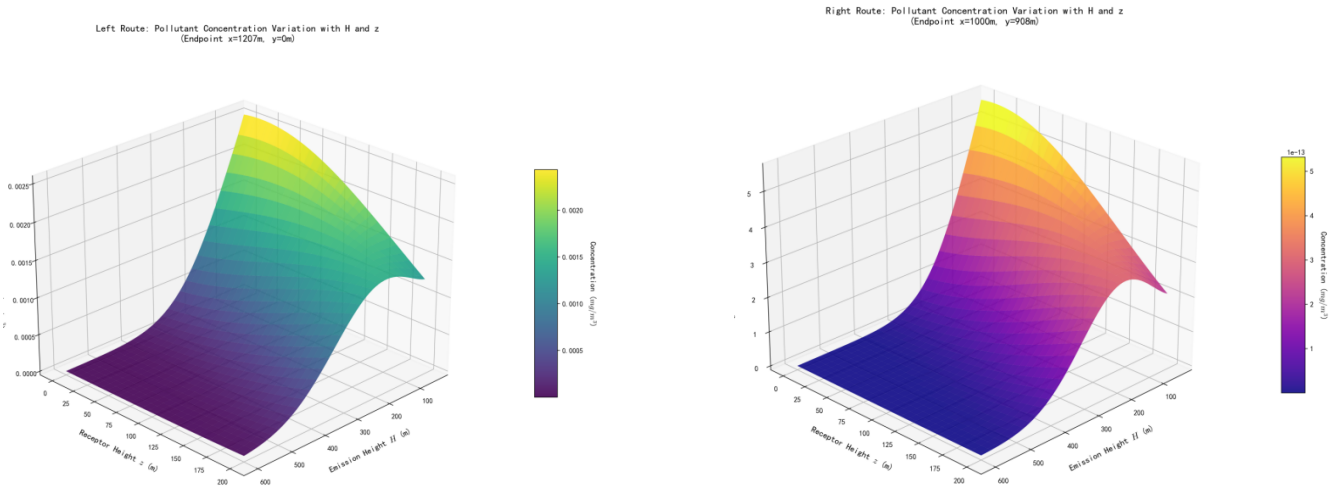


Figure 15: Three-dimensional concentration analysis of Guangzhou neighborhoods.

Three-dimensional surface analysis reaffirms the downwind axis as the primary target for emergency response. Proper emission height and receptor distance can significantly reduce environmental risks.

4. Problem 2 Model

4.1. Analysis of Pollutant Transport Mechanisms

Studying how outdoor pollutants enter indoor spaces requires understanding their transport in air, primarily governed by:

- **Diffusion:** Molecular motion along concentration gradients (Fick's law), causing slow spreading like ink in water.
- **Convection (Advection):** Macroscopic transport by airflow, the main driver of rapid pollutant movement.

In this model, three pathways through which outdoor pollutants enter indoor spaces are considered:

- **Door and window crack infiltration:** Slow natural infiltration through small gaps.
- **Natural ventilation:** Air exchange through open windows driven by wind or thermal pressure.
- **Mechanical ventilation:** Forced outdoor air supply via systems with exhaust outlets.

4.2. Construction of the Governing Equation

To describe the variation of pollutant concentration (C) with time (t) and spatial coordinates ((x, y, z)), a three-dimensional unsteady advection–diffusion equation is established as the core governing equation. The general form is as follows:

$$\frac{\partial C}{\partial t} + \nabla \cdot (\mathbf{u}C) = \nabla \cdot (D\nabla C) + S \tag{12}$$

where $\mathbf{u} = (u, v, w)$ is the velocity vector, D is the diffusion coefficient, and S is the source term.

Assuming that the indoor flow field is incompressible ($\nabla \cdot \mathbf{u} = 0$) and that there are no internal sources indoors ($S = 0$), the equation can be expanded and simplified as:

$$\frac{\partial C}{\partial t} = D\left(\frac{\partial^2 C}{\partial x^2} + \frac{\partial^2 C}{\partial y^2} + \frac{\partial^2 C}{\partial z^2}\right) - \left(u\frac{\partial C}{\partial x} + v\frac{\partial C}{\partial y} + w\frac{\partial C}{\partial z}\right) \tag{13}$$

The left-hand side of the equation represents the local rate of change of concentration. The first term on the right-hand side describes molecular diffusion, while the second term represents the transport of pollutants by the velocity field.

4.3. Boundary Conditions

To solve the above partial differential equation, appropriate boundary and initial conditions must be specified according to the physical scenario.

4.3.1. *Boundary Conditions*

The following boundary conditions are defined according to different types of ventilation openings:

1. **Door and window crack infiltration boundary (Γ_{crack}):**

   Considering the resistance of cracks to airflow, an infiltration coefficient (γ_{inf}) is introduced. The concentration at the boundary is proportional to the outdoor concentration (C_{out}):

$$C(x, y, z, t) = C_{\text{out}} \cdot \gamma_{\text{inf}}, \quad \forall (x, y, z) \in \Gamma_{\text{crack}}, \forall t > 0 \tag{14}$$

2. **Window natural/mechanical air supply boundary (Γ_{window}):**

   Assuming that open windows or mechanical air inlets are directly connected to the outdoors, the boundary concentration is equal to the outdoor pollutant concentration:

$$C(x, y, z, t) = C_{\text{out}}, \quad \forall (x, y, z) \in \Gamma_{\text{window}}, \forall t > 0 \tag{15}$$

3. **Mechanical exhaust outlet boundary (Γ_{exhaust}):**

   Assuming that the exhaust outlet is equipped with an air purification device or treated as an outflow boundary, the concentration is set to zero (absorbing boundary):

$$C(x, y, z, t) = 0, \quad \forall (x, y, z) \in \Gamma_{\text{exhaust}}, \forall t > 0 \tag{16}$$

4. **Wall boundary (Γ_{wall}):**

   Assuming that the walls are impermeable and pollutants cannot pass through them, a Neumann zero-flux boundary condition is imposed:

$$-D \cdot \frac{\partial C}{\partial \mathbf{n}}(x, y, z, t) = 0, \quad \forall (x, y, z) \in \Gamma_{\text{wall}}, \forall t > 0 \tag{17}$$

4.3.2. *Initial Condition*

It is assumed that the indoor air is clean at the initial moment, with no pollutant present:

$$C(x, y, z, 0) = 0, \quad \forall (x, y, z) \in \Omega \tag{18}$$

4.4. Numerical Solution Method

Since the governing equation is difficult to solve analytically under complex boundary conditions, the finite difference method (Finite Difference Method, FDM) is adopted for numerical discretization and solution.

- **Spatial discretization:** The three-dimensional room is divided into a regular grid system.
- **Temporal discretization:** Explicit or implicit time-stepping schemes are employed.
- **Stability treatment:** For the advection term, an upwind differencing scheme is used to suppress numerical oscillations and ensure the physical reliability of the results.

4.5. Simulation Scenario Setup

To systematically analyze pollutant diffusion under different ventilation conditions, a standard three-dimensional indoor space model is established.

- **Room geometry:** An ideal room model with dimensions $L_x = 5\,\text{m}$, $L_y = 4\,\text{m}$, $L_z = 3\,\text{m}$ is constructed.
- **Ventilation components:** The model includes door and window cracks (Crack), natural ventilation windows (Natural Vent. Window), mechanical air supply inlets (Mech. Supply), and mechanical exhaust outlets (Mech. Exhaust), as shown in the figure.
- **Eight combined operating conditions:** Based on the combinations of three variables—"whether cracks are permeable," "whether windows are open," and "whether the mechanical system is operating"—a total of eight simulation scenarios are constructed (ranging from a fully sealed baseline to the combined action of all three).

4.6. Model Solution and Results Analysis

By running the numerical model, the diffusion process of pollutants blown into the indoor environment is analyzed in depth, from a three-dimensional macroscopic perspective to a one-dimensional microscopic

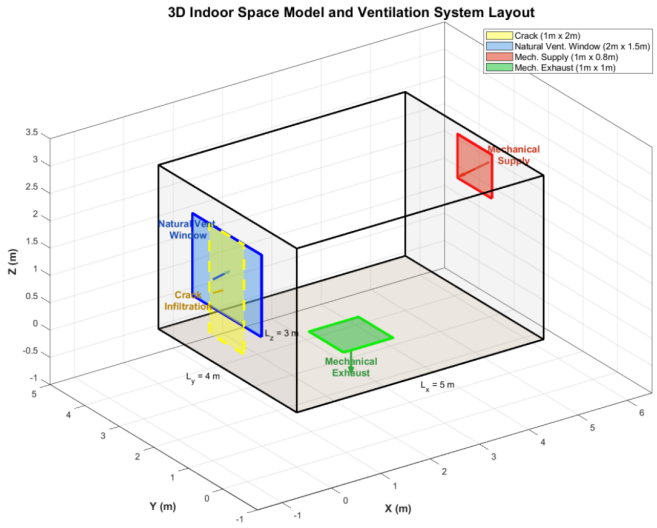


Figure 16: One-Dimensional Profile Waterfall Plot

perspective.

4.6.1. *Three-Dimensional Macroscopic Diffusion Process Analysis*

The following figure shows time-evolving slice plots of the indoor pollutant concentration field under the ”combined action of all three” operating condition.

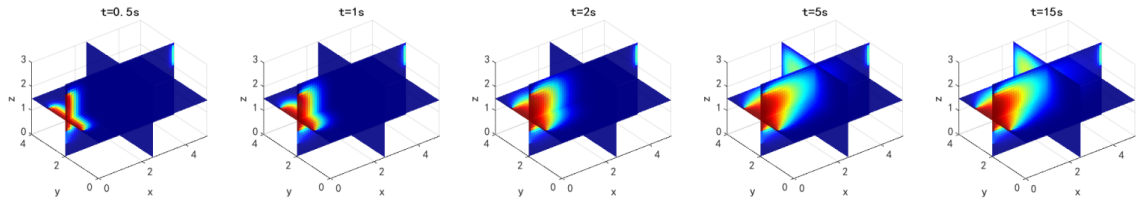


Figure 17: Temporal Evolution of Three-Dimensional Concentration Slices

Based on orthogonal slice analysis of the three-dimensional concentration field, the complex dynamics of pollutant transport are revealed: the high-concentration core region preferentially extends along the height of the air inlet and does not immediately settle, reflecting significant vertical concentration gradients and lag effects. Meanwhile, pollutant transport in the downwind direction (X-axis) is much faster than lateral and vertical transport. This anisotropic diffusion phenomenon directly confirms the dominant role of convection in the main flow direction, while mixing perpendicular to the flow is primarily limited by inefficient molecular diffusion.

4.6.2. *Two-Dimensional Cross-Sectional Multi-Scenario Comparative Analysis*

To compare the efficiency of different ventilation mechanisms, Figure 18 presents the two-dimensional concentration field distributions at four key time points under eight scenarios.

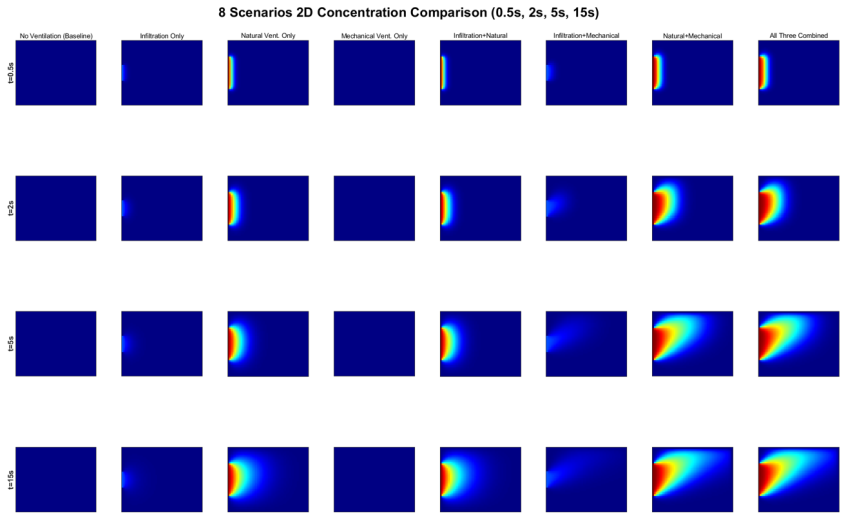


Figure 18: Comparative Two-Dimensional Concentration Distributions under Eight Scenarios

The evolution patterns of the two-dimensional concentration fields clearly distinguish diffusion-dominated and convection-dominated physical mechanisms. In infiltration scenarios lacking active ventilation, pollutants can only propagate slowly in an isotropic, arc-shaped manner characteristic of molecular diffusion, resulting in significantly limited transport efficiency. Once forced convection induced by mechanical or natural ventilation is introduced, the concentration field rapidly transforms into a parabolic distribution elongated along the flow direction. This strong directional mechanism dominated by advective transport greatly accelerates the migration of high-concentration outdoor pollutants into the interior of the room.

4.6.3. *One-Dimensional Waterfall Plot Analysis Along the Z-Axis*

To investigate vertical concentration stratification, a waterfall plot of concentration distribution along the Z-direction is constructed.

In the vertical concentration profiles under full ventilation conditions, the clearly identifiable wavefront propagation and gradual flattening process reveal the dominant role of convection in pushing the high-concentration core into the interior of the room and establishing a stable gradient. Meanwhile, the smooth and continuous characteristics of the evolving curves, without non-physical oscillations, strongly verify the numerical stability of the differencing scheme adopted in this model for handling advection–diffusion problems.

4.6.4. *Indoor Average Concentration and Dimensionless Analysis*

This figure quantitatively shows the variation trend of the indoor average concentration and the time required to reach specific concentration levels.

Dimensionless concentration analysis shows strong self-similarity, with normalized curves under different conditions following consistent exponential growth, indicating that pollutant transport in confined spaces obeys first-order linear dynamics. Mechanical ventilation reaches 90% of steady-state in under 20 seconds, while crack infiltration takes much longer, highlighting critical time-scale differences for sudden exposure risk assessment.

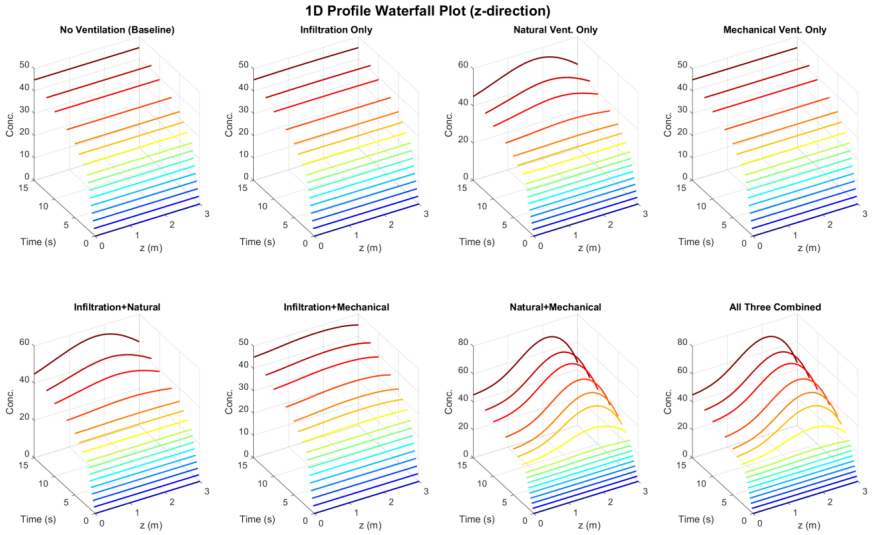


Figure 19: One-Dimensional Concentration Profile Waterfall Plot

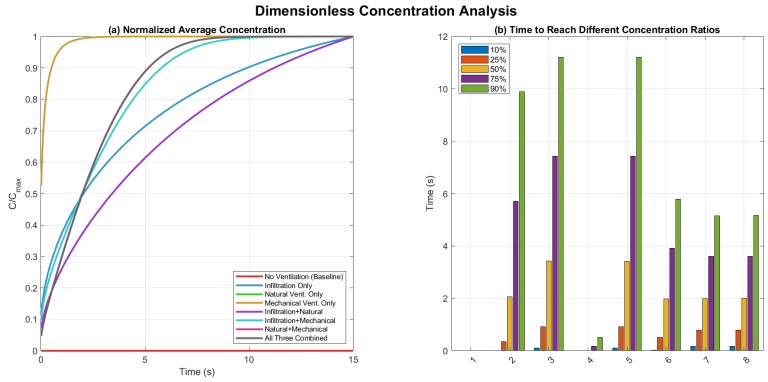


Figure 20: Analysis of Dimensionless Concentration and System Response Time

4.6.5. *Spatiotemporal Heatmap Evolution Analysis (Center Point)*

The geometric center of the room ($(x = 2.5, y = 2.0, z = 1.5)$) is selected. By fixing two coordinates and observing the concentration variation over time along the third coordinate axis, x-t, y-t, and z-t plots are obtained.

One-dimensional spatiotemporal heatmaps reveal distinct transport dynamics: along x-t, the inclined advancing front reflects convective dominance, with the wavefront slope (dx/dt) matching mean wind speed, confirming flow control. In contrast, y-t shows uniform mixing, while z-t exhibits stratification due to inlet geometry, with pollutants filling upper space first and slowly diffusing downward, creating a pronounced vertical time lag.

4.6.6. *Dynamic Response Analysis of Monitoring Points*

Finally, six representative locations within the room (such as near windows, the center, corners, and the ceiling) are selected as monitoring points to analyze the temporal evolution of concentration.

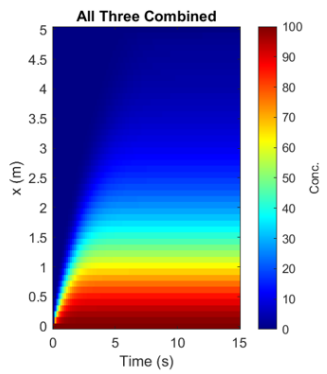


Figure 21: x-t plot

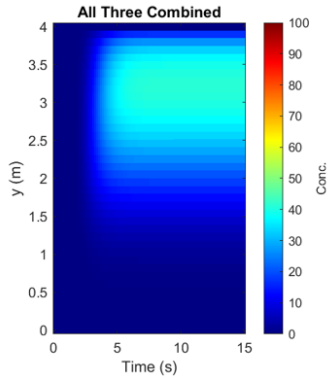


Figure 22: y-t plot

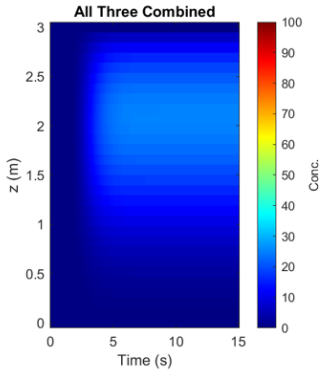


Figure 23: z-t plot

Figure 24: One-dimensional spatiotemporal plots

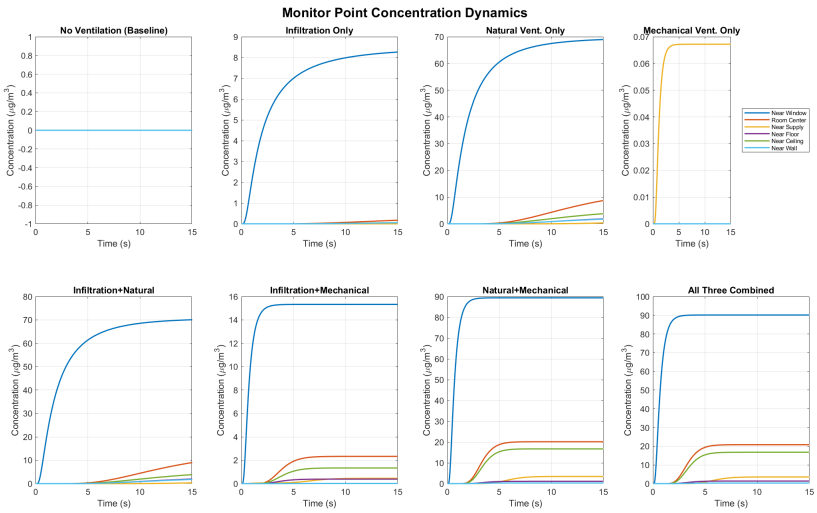


Figure 25: Dynamic concentration variations at monitoring points under different scenarios

Dynamic analysis of monitoring points shows strong spatial heterogeneity: concentrations near inflow boundaries peak first, while the room center and corners exhibit pronounced time-lag effects. At the breathing-zone height ($z = 1.5\,\mathrm{m}$), mechanical ventilation quickly drives relative concentrations above 80% of outdoor levels within seconds, indicating a very limited indoor response window during sudden pollution events.

4.7. Practical Implications

Based on the above model analysis, conclusions with important practical implications can be drawn:

- **Emergency response to sudden pollution events:** When sudden outdoor pollution occurs (such as chemical plant leaks or terrorist attacks), model results show that mechanical ventilation can cause indoor concentrations to reach hazardous levels within 20 seconds. Therefore, immediately shutting down mechanical ventilation systems and closing windows is the most critical measure to reduce indoor occupant exposure risk.

- **Shelter strategy selection:** Due to the pronounced "time lag" effect of crack infiltration (requiring a long time to reach high concentrations), when immediate evacuation is not possible, seeking well-sealed rooms (reducing crack infiltration γ_{inf}) for shelter-in-place is a better option than blindly rushing outdoors.
- **Implications for building design:** The relative positions of air supply and exhaust outlets directly affect vertical stratification. To improve ventilation efficiency under normal conditions, airflow paths should be designed to exploit convection-dominated characteristics. However, when defending against external pollution, building airtightness (reducing unorganized Γ_{crack}) is crucial.

5. Model Evaluation

5.1. Strengths

- **Environmental adaptability and boundary handling:** The model uses the **Virtual Source Method** to build a segment-coupled Gaussian model, smoothing concentration variations over heterogeneous urban surfaces, and automatically switches to a **Gaussian Puff Model** in calm winds, avoiding low-speed singularities and enhancing applicability.
- **Macro–micro scale coupling:** The framework simulates **urban-to-indoor dispersion**, with a 3D advection–diffusion model incorporating ventilation, crack infiltration, deposition, and wall reflection to quantify pollutant anisotropy and time-lag effects.

5.2. Weaknesses

- **Flow field simplification:** The outdoor model, based on time-averaged Gaussian assumptions, simplifies micro-meteorology and cannot capture transient turbulence or local vortices in dense urban "street canyons."
- **Idealized geometry and parameters:** The indoor model assumes unobstructed spaces, ignoring furniture, occupants, and heat sources, and relies on empirical parameters (e.g., Briggs coefficients, ground reflection), making predictions uncertain under extreme conditions without calibration.

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Report on Use of AI Tools

1.Tool Name: Doubao
Version Information: v2025 Official Release (Update Date: January 2025)
Purpose of Use: Paper Translation
Notes: The translated paper has undergone multiple rounds of rigorous verification, with numerous semantic deviations and logical errors corrected. The final version meets the required academic standards for papers.

2. Tool Name: Doubao
Version Information: v2025 Official Release (Update Date: January 2025)
Purpose of Use: Assisting in Literature, Formula, and Empirical Parameter Retrieval

Question one
Provide the derivation process of the Gaussian plume model.
Anser

The Gaussian plume model is a classic atmospheric pollutant diffusion model based on **turbulent diffusion theory**. Its core is derived from the **three-dimensional steady-state advection-diffusion equation** combined with the **normal distribution assumption** and the **image source method**. It is applicable to continuous point source emission scenarios under flat terrain and steady wind fields.

1. Model Assumptions

1. **Steady-state condition**: The wind speed u is constant, the source strength Q is stable, and the concentration field does not change with time, i.e., $\frac{\partial C}{\partial t} = 0$.
2. **Coordinate system definition**: The wind direction is along the x-axis. Only turbulent diffusion in the y (horizontal crosswind) and z (vertical) directions is considered, while diffusion in the x-direction is neglected (since advection dominates).
3. **Normal distribution hypothesis**: The pollutant concentration follows a normal distribution in the y and z directions, and the turbulent diffusion coefficients D_y and D_z are constants.
4. **Ground boundary condition**: The ground surface ($z = 0$) is a perfect reflector with no absorption or sedimentation. The image source method is used to handle this boundary condition.
5. **Inert pollutant assumption**: The pollutant is an inert substance with no mass loss due to chemical reactions, sedimentation, evaporation, etc.

2. Establishment of the Governing Equation

Pollutant transport satisfies the **three-dimensional steady-state advection-diffusion equation** (mass conservation), which is simplified based on the above assumptions:

1. **Original equation**:
   $u\frac{\partial C}{\partial x} = D_y\frac{\partial^2 C}{\partial y^2} + D_z\frac{\partial^2 C}{\partial z^2}$
   where $C(x, y, z)$ is the pollutant concentration; u is the average wind speed along the x-axis; D_y and D_z are the turbulent diffusion coefficients in the y and z directions, respectively.

3. Solution for Unbounded Space (Free Atmosphere)

1. **Normal distribution form assumption**: The concentrations in the y and z directions follow independent normal distributions, expressed as:

$$C(x,y,z) = A(x)\exp\left(-\frac{y^2}{2\sigma_y^2(x)}\right)\exp\left(-\frac{z^2}{2\sigma_z^2(x)}\right)$$

where $A(x)$ is the concentration amplitude function in the x-direction; $\sigma_y(x)$ and $\sigma_z(x)$ are the diffusion standard deviations in the y and z directions, respectively, and $\sigma_y^2 = 2D_y x/u$, $\sigma_z^2 = 2D_z x/u$ (derived from turbulent diffusion theory).

2. **Source strength constraint**: The source strength conservation condition $Q = \int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} uC\ dydz$ must be satisfied through full-space integration. Substitute the normal distribution form into the integral to obtain:

$$A(x) = \frac{Q}{2\pi u\sigma_y\sigma_z}$$

3. **Unbounded space solution**:

$$C(x,y,z) = \frac{Q}{2\pi u\sigma_y\sigma_z}\exp\left(-\frac{y^2}{2\sigma_y^2}\right)\exp\left(-\frac{z^2}{2\sigma_z^2}\right)$$

4. Incorporating Ground Reflection (Image Source Method)

1. **Boundary condition**: No pollutant flux across the ground surface ($z = 0$), i.e., $\frac{\partial C}{\partial z}\big|_{z=0} = 0$.
2. **Image source method**: To satisfy the zero-flux boundary condition, a virtual image source with the same strength as the real source is placed at $z = -H$ (symmetric to the real source at $z = H$). The actual concentration field is the superposition of the concentrations from the real source and the image source.
3. **Elevated point source solution**:

$$C(x,y,z) = \frac{Q}{2\pi u\sigma_y\sigma_z}\exp\left(-\frac{y^2}{2\sigma_y^2}\right)\left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)\right]$$

where H is the **effective stack height** (sum of the physical stack height and the plume rise height).

5. Parameter Description

Parameter	Meaning	Unit
$C(x,y,z)$	Pollutant concentration at spatial point (x,y,z)	$\mathrm{kg/m^3}$ or $\mathrm{g/m^3}$
Q	Source strength (pollutant emission rate)	$\mathrm{kg/s}$ or $\mathrm{g/s}$
u	Average wind speed	$\mathrm{m/s}$
σ_y, σ_z	Diffusion standard deviations in y and z directions (increasing with downwind distance x)	m
H	Effective emission height	m

6. Summary of Key Derivation Steps

1. Establish the concentration distribution form based on steady-state and normal distribution assumptions.
2. Determine the amplitude function $A(x)$ using the source strength conservation constraint.
3. Handle the ground boundary condition with the image source method and superimpose the concentrations from the real source and the image source.
4. Derive the final Gaussian plume model formula for elevated continuous point sources.

Question two

Air pollution indices, calculation formulas for diffusion coefficients, and different reflection coefficients corresponding to underlying surfaces.
Anser

1. Air Pollution Indices

Air pollution indices (APIs) are standardized metrics used to quantify the degree of air pollution and its impact on human health and the environment. Common indices include:

- **Air Quality Index (AQI)**: A universal index that integrates the concentrations of key pollutants (e.g., $PM_{2.5}$, PM_{10}, SO_2, NO_2, CO, O_3). It is calculated by normalizing the concentration of each pollutant to a reference standard and selecting the maximum normalized value as the AQI of the region. The formula for a single pollutant sub-index is:$I_p = \frac{I_{p,high} - I_{p,low}}{C_{p,high} - C_{p,low}} \times (C_p - C_{p,low}) + I_{p,low}$Where:
  - I_p = Sub-index of pollutant p;
  - C_p = Measured concentration of pollutant p;
  - $C_{p,high}, C_{p,low}$ = Upper and lower limit concentrations of the pollution level interval where C_p falls;
  - $I_{p,high}, I_{p,low}$ = Upper and lower limit values of the sub-index corresponding to this interval.
- **Pollutant Standard Index (PSI)**: Widely used in some regions, it focuses on six pollutants and classifies air quality into 6 levels, ranging from "good" to "hazardous".

2. Calculation Formulas for Diffusion Coefficients

In atmospheric diffusion models (e.g., Gaussian plume model), **turbulent diffusion coefficients** (D_y for horizontal direction, D_z for vertical direction) describe the ability of turbulent eddies to transport pollutants. Two main calculation methods are:

(1) Based on Diffusion Standard Deviations (σ_y, σ_z)

From the core relationship between diffusion coefficients and diffusion standard deviations in turbulent diffusion theory:

$$\sigma_y^2 = 2D_y\frac{x}{u}, \quad \sigma_z^2 = 2D_z\frac{x}{u}$$

Rearranged to get the diffusion coefficient formulas:

$$D_y = \frac{u\sigma_y^2}{2x}, \quad D_z = \frac{u\sigma_z^2}{2x}$$

Where:

- u = Average wind speed along the downwind direction (m/s);
- x = Downwind distance from the pollution source (m);
- σ_y, σ_z = Horizontal and vertical diffusion standard deviations (m), which are empirical values related to atmospheric stability and downwind distance (can be queried from Pasquill-Gifford curves).

(2) Empirical Formulas for Atmospheric Stability Classes

Atmospheric stability is divided into 6 classes (A-F, from extremely unstable to extremely stable). For different stability classes, empirical formulas for σ_y and σ_z are used first, then substituted into the above equations to calculate D_y and D_z. For example:

- For **Pasquill stability class C** (neutral stability) and $x < 1000\ \mathrm{m}$:$\sigma_y = 0.15x, \quad \sigma_z = 0.10x$

3. Reflection Coefficients Corresponding to Different Underlying Surfaces

The **ground reflection coefficient** (γ) describes the ratio of pollutant flux reflected by the underlying surface to the incident flux, which is used to modify the Gaussian plume model when the ground is not a perfect reflector (i.e., $\gamma \neq 1$). Its value depends on the physical properties of the underlying surface, and typical values are shown in the table below:

Underlying Surface Type	Reflection Coefficient (γ)	Physical Explanation
Smooth Water Surface (Lake, Sea)	0.90-1.00	Almost perfect reflection; minimal absor deposition of pollutants
Dense Forest	0.10-0.30	Strong adsorption and deposition effect vegetation canopy; most pollutants are r
Urban Built-up Area	0.40-0.60	Complex surface structure; partial reflect deposition on buildings and roads
Farmland/Cropland	0.50-0.70	Moderate retention capacity; influenced coverage
Bare Soil/Gravel Land	0.60-0.80	Weak adsorption capacity; higher reflect vegetation-covered surfaces

When $\gamma < 1$, the modified Gaussian plume model for elevated sources is:

$$C(x,y,z) = \frac{Q}{2\pi u \sigma_y \sigma_z} \exp\left(-\frac{y^2}{2\sigma_y^2}\right) \left[\exp\left(-\frac{(z-H)^2}{2\sigma_z^2}\right) + \gamma \exp\left(-\frac{(z+H)^2}{2\sigma_z^2}\right)\right]$$

Notes:All retrieved literatures and empirical parameters have been properly cited with corresponding references.

3. Tool Name: Deepseek
Version Information: Deepseek-v3.2
Purpose of Use: Assisting in Code Development
Inquiry in Question 2: Developing code for solving the advection-diffusion model using the finite difference method.